SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (22 marks – approximately 40 minutes)

A key relationship in macroeconomics, especially from an economic policy perspective, is the relationship between changes in spending and changes in real GDP (real income). If an economy has room to expand, so that increase in spending does not lead to increase in price, then there is a direct relationship between the two aggregates. For example, an increase in investment would eventually result in more than a proportionate increase in GDP, and vice versa.

Required:

(a) **State the general formula for the "expenditure approach" used in measuring an open economy's total output.**

(2 marks)

(b) Consider a closed economy with no tax imposed by the government and assume increases in spending do not lead to an increase in inflation rate. Suppose the investment in an economy increases by $10 billion and GDP goes up by $50 billion.

(i) **Compute the multiplier, marginal propensity to consume (MPC) and marginal propensity to save (MPS).**

(5 marks)

(ii) **Determine the missing figures in the following table based on the results in part (b)(i).**

	(1) Change in Income (in billion)	(2) Change in Consumption (in billion) MPC (=_____)	(3) Change in Saving (in billion) MPS (= _____)
Increase in investment of $10 billion	$10		
Second round			
Third round			
All other rounds			
Total:	$50		

(5 marks)

(iii) Explain how the multiplier effect is affected by an increase in

- **marginal propensity to consume; and**
- **marginal propensity to save.**

(3 marks)

(iv) Describe any TWO factors that influence an individual's marginal propensity to consume.

(4 marks)

(v) Analyse how the multiplier effect calculated above would be affected if the economy highlighted above is an open economy with government tax.

(3 marks)

<u>Question 2</u> (24 marks – approximately 43 minutes)

(a) Analyse the following statement and justify whether it is true or false.

"All else the same, a decrease in the supply of RAM chips for computers would result in a shortage of RAM chips."

(5 marks)

(b) Nashille Co, an Italian pasta boutique, produces an exceptional hand-made ravioli, a kind of Italian pasta. Last month, the company sold 5,000 packs of ravioli at a price of HK$100 each.

<u>Required:</u>

(i) Compute the monthly quantity demanded of the product if the firm lowers the price by HK$10 and the (relevant) price elasticity of demand over this range is -2.

(5 marks)

(ii) Compute the change in revenue because of the price change in part (b)(i).

(3 mark)

(iii) Without doing any calculations, explain how the total revenue will change if the price elasticity of demand is -1 over the range (instead of -2.0).

(3 marks)

(c) Suppose the poultry market (such as that of chicken) is a perfectly competitive market and in a state of long run equilibrium.

<u>Required:</u>

(i) Back in early 2000s, the fear of bird flu significantly affected the demand for chicken. Explain how this change in demand would affect, in the short run,

- **the equilibrium price and quantity; and**
- **the output of individual farmers.**

(3 marks)

(ii) Suppose the change in demand mentioned in part (c)(i) is permanent. Explain the long-run effects on the equilibrium price and quantity in the market, and the output of individual farmers.

(5 marks)

<u>Question 3</u> (9 marks – approximately 16 minutes)

Suppose a market research firm is hired to estimate the percentage of households living in a city which own a made-in-Japan flat screen TV. Of the 1,500 households sampled, 580 replied "yes".

<u>Required:</u>

(a) Define the confidence interval.

(2 marks)

(b) Compute the 95% confidence interval for the proportion of households in the city that claim to own a made-in-Japan flat screen TV.

(4 marks)

(c) Interpret the confidence interval estimate calculated in part (b).

(2 marks)

(d) Without doing any calculations, state (no explanations required) whether a 90% confidence interval for the proportion would be wider than, narrower than, or the same as the interval found in part (b).

(1 mark)

Question 4 **(25 marks – approximately 45 minutes)**

You work as a junior equity analyst in a local brokerage firm. Your latest assignment is to analyse the stock performance of a 4-year old hypothetical start-up company, TROOL Inc. The table below shows the quarterly nominal rate of returns (arranged in ascending order) for the common stock of the firm for the last four years.

Quarterly Percentage Returns for TROOL Inc. Common Stock															
-7	-6	-5	-4	-3	-2	-1	0	0	1	2	3	4	5	6	7

Required:

(a) **Explain the difference between a parameter and a sample statistic.**

(3 marks)

(b) **State TWO examples of a parameter.**

(2 marks)

(c) **Compute the range and the arithmetic average of the quarterly returns.**

(4 marks)

(d) **Define median and mode and compute their values based on the data in the table.**

(4 marks)

(e) **Based on the results in part (c) and (d), describe the position and shape of the return distribution.**

(4 marks)

(f) **Compute the standard deviation of the quarterly returns.**

(4 marks)

(g) **Suppose the nominal rates of return in the table are converted into real rates of return by subtracting the inflation rate (assumed to be 1%).**

Without doing any calculations, explain how the numerical value of the mean and standard deviation would be affected.

(4 marks)

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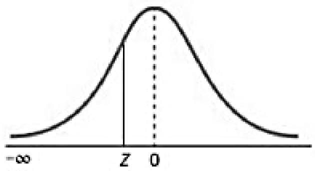
APPENDIX

The Cumulative Standardised Normal Distribution



Entry represents area under the cummulative standardised normal distribution from −∞ to Z

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
−3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
−3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
−3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
−3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
−3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
−3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
−3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
−3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
−3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00103	0.00100
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2388	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2482	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641